

• Sometimes $\mathcal{L}_\xi \mathcal{L} = 0 \Rightarrow \nabla_\mu \xi^\mu = 0$, solution
 $\xi^\mu = a^\mu + b^{\mu\nu} x_\nu$ (Pohorec)

$$\delta \mathcal{L} = \delta \Phi^i \frac{\delta \mathcal{L}}{\delta \Phi^i} + d\Theta[\delta \Phi, \Phi]$$

• Boundary term (Presymplectic form).

$\Theta \equiv \Theta^{(n-1)}$: $n-1$ form

$\mathcal{L} \equiv \mathcal{L}^{(n)}$: n -form

$\rightarrow (d\Theta)^{(n)}$

$$\delta \mathcal{L} = \delta \Phi^i \frac{\delta \mathcal{L}}{\delta \Phi^i} + \partial_\mu (\delta \Phi^i) \frac{\delta \mathcal{L}}{\delta (\partial_\mu \Phi^i)} + \dots = \delta \Phi^i \frac{\delta \mathcal{L}}{\delta \Phi^i} + d\Theta^a$$

$$d\Theta = dx^\nu \partial_\nu (\Theta^\mu (d^{n-1}x)_\mu) = \partial_\nu \Theta^\mu (d^n x) \delta_\mu^\nu = \partial_\mu \Theta^\mu d^n x$$

* Take the variation of $\mathcal{L}$ generated by an int. diffeo generated by ξ^μ

$$\delta_\xi \mathcal{L} = \delta_\xi g_{\mu\nu} \frac{\delta \mathcal{L}}{\delta g_{\mu\nu}} + \delta_\xi \Phi^i_\mu \frac{\delta \mathcal{L}}{\delta \Phi^i_\mu} + d(\dots)$$

$$= \mathcal{L}_\xi g_{\mu\nu} \frac{\delta \mathcal{L}}{\delta g_{\mu\nu}} + \delta_\xi \Phi^i_\mu \frac{\delta \mathcal{L}}{\delta \Phi^i_\mu} + d(\dots)$$

$$= 2 \nabla_\mu \xi^\mu \mathcal{L} / \delta g_{\mu\nu} + \delta_\xi \Phi^i_\mu \delta \mathcal{L} / \delta \Phi^i_\mu + d(\dots)$$

$$= 2 \nabla_\mu \xi^\mu \left(\frac{\sqrt{-g}}{2} \right) T^{\mu\nu} + \delta_\xi \Phi^i_\mu \delta \mathcal{L} / \delta \Phi^i_\mu + d(\dots)$$

• Note $\delta_\xi \Phi^i_\mu = \mathcal{L}_\xi \Phi^i_\mu$ and $\frac{\delta \mathcal{L}}{\delta \Phi^i_\mu} = d^n x \frac{\delta \mathcal{L}}{\delta \Phi^i_\mu}$

• Can build $\boxed{\mathcal{J}^\mu \equiv T^{\mu\nu} \xi_\nu}$ s.c.

$$\nabla_\mu \mathcal{J}^\mu = 0 \quad \text{or}$$

$\bar{\mathcal{J}} \equiv \mathcal{J}^\mu \sqrt{-g} (d^{n-1}x)_\mu$ is closed with $\ast \bar{\mathcal{J}}$

$$\nabla_\mu \mathcal{J}^\mu = \nabla_\mu T^{\mu\nu} \xi_\nu + T^{\mu\nu} \nabla_\mu \xi_\nu = 0 + 0 \Rightarrow d\bar{\mathcal{J}} = 0$$

Def: A Cauchy Surface is a subset of $\mathcal{M}$ which is intersected by every maximal causal curve exactly once.

$$Q \equiv \int_\Sigma \bar{\mathcal{J}}$$

which is conserved when the system evolves provided that no flux escapes sufficiently rapidly at the boundary.

$\partial \Sigma$.

* $x^0 = t$ at $t=0$, $(d^{n-1}x)_0 = d\Sigma$

$$\partial_t Q = \partial_t \int_{\Sigma} J^0 (d^{n-1}x)_0 = \int_{\Sigma} \partial_0 J^0 d\Sigma$$

$$\nabla_{\mu} J^{\mu} = \partial_0 J^0 + \nabla \cdot \vec{J} = 0$$

$$\Rightarrow \partial_t Q = - \int_{\Sigma} \nabla \cdot \vec{J} d\Sigma = - \int_{\partial \Sigma} \vec{J} \cdot d\vec{S} = 0$$

Covariant Phase space: linear functions on T^*M

$i: T_p M \xrightarrow{\sim} K(T_p^* M)$

$\xi \mapsto [i_{\xi}: T_p^* M \rightarrow \mathbb{R} \mid \omega \mapsto i_{\xi} \omega = (\xi^{\mu} \partial_{\mu}) \omega]$

$$i_{\xi} \omega = \xi^{\mu} \partial_{\mu} (\omega_{\sigma} dx^{\sigma}) = \omega_{\sigma} dx^{\sigma} (\xi^{\mu} \partial_{\mu})$$

$$= \xi^{\mu} \omega_{\mu} ?$$

For any $\omega \in \Omega^k$ we have $i_{\xi}: \Omega^k(M) \rightarrow \Omega^{k-1}(M)$

$i_{\xi} \omega \in \Omega^{k-1} \mid (i_{\xi} \omega)(x_1, \dots, x_{k-1}) = \omega(\xi, x_1, \dots, x_{k-1})$

• Also have the operator $d = dx^{\mu} \otimes \partial_{\mu}$ which acts on $T_p^* M \otimes T_p M$

$$\Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \Omega^2(M) \rightarrow \dots$$

$$\xleftarrow{i_{\xi}} \quad \xleftarrow{i_{\xi}}$$

* $\Phi \equiv (\Phi^i)_{i \in I} \ni g_{\mu\nu}, \phi_i, A_{\mu} \in \mathfrak{g} \dots$

$\text{Span} \{ \Phi^i, \Phi^i_{,\mu}, \Phi^i_{,\mu\nu}, \dots \} = \mathcal{J}$

$\searrow \partial_{\mu} \partial_{\nu} \Phi^i \dots$

A "point" on the jet space is $(\Phi^i, \Phi^i_{,\mu}, \Phi^i_{,\mu\nu}, \dots)$

$$T^* \mathcal{J} = \text{span} (\delta \Phi^i, \delta \Phi^i_{,\mu}, \delta \Phi^i_{,\mu\nu}, \dots)$$

The symmetrized derivative is defined as

$$\frac{\partial \Phi^i_{\alpha\beta}}{\partial x^{\gamma}} = \delta^{\alpha}_{\gamma} \delta^{\beta}_{\mu} \delta^i_{,\mu} = \frac{1}{2} (\delta^{\alpha}_{\gamma} \delta^{\beta}_{\mu} + \delta^{\beta}_{\gamma} \delta^{\alpha}_{\mu}) \delta^i_{,\mu}$$

$$\frac{\partial \Phi^i_{,\mu\nu}}{\partial x^{\gamma}}$$

Ex: $\frac{\partial \Phi^i_{,\mu\nu}}{\partial x^{\gamma}} = \frac{1}{2} (\delta^{\mu}_{\gamma} \delta^{\nu}_{\rho} + \delta^{\nu}_{\gamma} \delta^{\mu}_{\rho}) \delta^i_{,\rho}$

$\frac{\partial \Phi^i_{,\mu\nu}}{\partial x^{\gamma}} = \frac{1}{2} \delta^i_{,\gamma}$

• δ is defined as

$$S \equiv \delta \Phi^i \frac{\partial}{\partial \Phi^i} + \delta \Phi^i_{,\mu} \frac{\partial}{\partial \Phi^i_{,\mu}} + \delta \Phi^i_{,\mu\nu} \frac{\partial}{\partial \Phi^i_{,\mu\nu}} + \dots$$

* Convention: All $\delta \Phi^i, \delta \Phi^i_{,\mu}$ are Grassmann ODD
 $\Rightarrow \delta^2 = 0$

Jet bundle: $(x^\mu, \Phi^i_{,\mu})$

• A section is of the form

$$\{\Phi^i(x^\mu), \partial_\mu \Phi^i(x^\mu), \partial_\mu \partial_\nu \Phi^i(x^\mu), \dots\}$$

• Differential operator: $d = dx^\mu \otimes \partial_\mu$ with

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} + \Phi^i_{,\mu} \frac{\partial}{\partial \Phi^i} + \Phi^i_{,\mu\nu} \frac{\partial}{\partial \Phi^i_{,\nu}} + \dots$$

• Horizontal ext. derivative $d := dx^\mu \wedge \partial_\mu$

$$\text{with } D_\mu = \partial_\mu + \delta \Phi^i_{,\mu} \frac{\partial}{\partial \Phi^i} + \delta \Phi^i_{,\mu\nu} \frac{\partial}{\partial \Phi^i_{,\nu}} + \dots$$

$$d \Phi^i_I = dx^\mu \wedge D_\mu \Phi^i_I = dx^\mu \wedge (\partial_\mu \Phi^i_I + \delta \Phi^i_{I\mu})$$

$$\Omega^{p,q}(\mathcal{J}^\infty E) \quad p: dx^\mu \quad q: \delta \Phi^i_I$$

$$\Omega^{p,q}(\mathcal{J}^\infty E) \ni \alpha = \alpha_{\mu_1 \dots \mu_p} \wedge_{i_1 \dots i_q} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p} \wedge \delta \Phi^{i_1}_{I_1} \wedge \dots \wedge \delta \Phi^{i_q}_{I_q}$$

$$d: \Omega^{p,q} \rightarrow \Omega^{p+1,q}$$

$$\delta: \Omega^{p,q} \rightarrow \Omega^{p,q+1}$$

* Note that

$$(a) \quad dx^\mu \wedge \delta \Phi^A = - \delta \Phi^A \wedge dx^\mu$$

$$(b) \quad \delta x^\mu \stackrel{!}{=} 0 = \delta(dx^\mu) = 0$$

$$(\delta d + d \delta) x^\mu = \delta(dx^\mu) + d(\delta x^\mu) = 0$$

• On a field: $d \Phi^i_I = dx^\mu \Phi^i_{I\mu}$

$$d \Phi^i_I = dx^\mu \Phi^i_{I\mu}$$

$$\delta(d \Phi^i_I) = \delta(dx^\mu \Phi^i_{I\mu}) = \delta dx^\mu \Phi^i_{I\mu} - dx^\mu \wedge \delta \Phi^i_{I\mu} \\ = - dx^\mu \wedge \delta \Phi^i_{I\mu}$$

$$d(\delta\Phi_I^i) = dx^\mu \wedge D_\mu(\delta\Phi_I^i)$$

$$D_\mu \delta\Phi_I^i = \delta\Phi_{I\mu}^i$$

so

$$d\delta\Phi_I^i = dx^\mu \wedge \delta\Phi_{I\mu}^i$$

* ~~Relation to Maxwell Action:~~

$$A = A_\mu dx^\mu, F = dA$$

$$(\delta_{\text{var}} F) = \partial_\mu (\delta_{\text{var}} A_\nu) - \partial_\nu (\delta_{\text{var}} A_\mu)$$

$$\text{or } \delta_{\text{var}} F = d(\delta_{\text{var}} A)$$

$$\text{but in } T^*\mathcal{M}: \delta(dA) = -d(\delta A)$$

$$\leadsto \delta A = -\delta_{\text{var}} A$$

$$\delta \mathbb{L} = \delta\Phi^i \frac{\partial \mathbb{L}}{\partial \Phi^i} - d\Theta[\delta\Phi, \Phi]$$

* Here we have the above defined for $\mathbb{L} \in \mathcal{L}(\mathcal{M})$
Contract each side with the interior derivative

$$i_{\delta_a} := \delta_a \Phi_I^i \frac{\partial}{\partial \Phi_I^i}$$

$\delta_a \Phi_I^i$: Grassmann
EVEN

$$\begin{aligned} i_{\delta_a}(\delta \mathbb{L}) &= \delta_a \Phi_I^i \frac{\partial}{\partial \Phi_I^i} \left(\delta\Phi_L^i \frac{\partial \mathbb{L}}{\partial \Phi_L^i} - d\Theta[\delta\Phi, \Phi] \right) \\ &= \delta_a \Phi_I^i \frac{\partial \mathbb{L}}{\partial \Phi_L^i} - \delta_a \Phi_L^i \frac{\partial}{\partial \Phi_L^i} \left[\partial_\mu \Theta_{\mu, \mu_1 \dots \mu_N} dx^{\mu_1} \dots \right] \\ &= \delta_a \Phi_I^i \frac{\partial \mathbb{L}}{\partial \Phi_L^i} \end{aligned}$$

Witten 98' (Covariant Paper)

$$T^*\mathcal{M}: \omega = dp^i \wedge dq_i$$

$$Q^{\mathbb{I}} = (p_i, q_i), \quad Q^i = p^i, \quad i \leq N, \quad Q^i = q^{i-N}, \quad i > N$$

$$A(Q^{\mathbb{I}}), B(Q^{\mathbb{I}}): [A, B] = \omega^{\mathbb{I}\mathbb{J}} \frac{\partial A}{\partial Q^{\mathbb{I}}} \frac{\partial B}{\partial Q^{\mathbb{J}}}$$

• Non degeneracy of ω : $\forall p \in T^*M$, $\omega_p(p)$ is invertible
 $\Rightarrow$ Do not need (p, q) but rather only need ω
 which is a non-degenerate closed 2-form

Def: Classical phase space: in a given physical theory, classical phase space is the space of solutions of the classical E.O.M.

• Ex: $L = \int_M (\frac{1}{2} \partial_\alpha \phi \partial^\alpha \phi - V(\phi))$

Q: How do we find a covariant description of a symplectic

structure of $J^\infty M$ of classical solⁿ?

$\rightarrow \Delta \phi \in T^\infty M: (x, \phi, \phi', \phi'', \dots)$

$0 = \Box \phi + V'(\phi)$

(i) $x \in M$, if $\phi \in \mathcal{S}$ then $\phi: M \rightarrow \mathbb{R}$

$\phi \in \Gamma(J^\infty M)$

(ii) A tangent vector: displacement in ϕ which preserves

E.O.M. & Requiring $\tilde{\phi} = \phi + \delta\phi$ obey E.O.M.

to the lowest order in $\delta\phi$

$\Box \tilde{\phi} + V'(\tilde{\phi}) = 0$

$\Box \phi + \Box \delta\phi + V'(\phi) + \delta\phi V''(\phi) = 0$

and so

$\Box \delta\phi + \delta\phi V''(\phi)$

